



CLIFF CASEY

SOFTWARE ARCHITECT

CONTACT



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Baltimore, MD

PROFILE

Software engineer with decades of experience building and scaling complex enterprise systems, steering technical decisionmaking across organizations, and solving difficult software architecture and engineering problems in high-volume and high-risk environments.

Led engineering organizations through technical guidance, coaching, code reviews, and sustainable engineering practices. Contributed extensively to international software engineering standards through work with the IEEE and other standards bodies, and to open source software projects directly. Guided responsible integration of LLM use into developer workflows, increasing productivity without sacrificing product quality or increasing technical debt.

SKILLS & TECHNICAL SUMMARY

Core Languages

Java, Js/Ts, Clojure, Python, PHP, C/C++, C#

Persistence

PostgreSQL, SQL Server, MySQL, SQLite, Graph & Document DBs, GraphQL, K-V Stores, Data Warehousing

Messaging

RabbitMQ, ActiveMQ, Kafka, SQS

Frontend

React (+Native), HTML5, jQuery, AngularJS, CSS/SASS

Machine Learning

LLM Code Analysis and Generation (Codex, Copilot), AI Policy, ML Classification and Regression Models

Containerization & IaC

Docker, Kubernetes, CloudFormation, ArgoCD

Cloud Infrastructure

AWS, Azure, PlatformOne

Collaboration Tools

Git, Jira, Asana, SVN, Confluence

CI/CD

GitHub Actions, Jenkins, Bamboo, BitBucket Pipelines

Game Engines

Unreal Engine 5 (C++ & BP), Unity, LibGDX

Security

RBAC, LDAP/AD, SSO/SAML, OAuth 2.0, TLS, Secure SDLC practices, PCI, SOC 2, NIST CSF & 800-171, OWASP

Protocols & Formats

JSON, REST, XML / SOAP, YAML, EDN

PROFESSIONAL EXPERIENCE

Chief Technology Officer

Yet Analytics Inc, Baltimore MD | Sep 2019-Present

Technical lead for an education technology software company delivering best-in-class learning data solutions to commercial and government clients. Responsibilities include architecture and engineering of software and cloud infrastructure, hiring and managing technical resources, managing client technical projects, and directing and advising the company from an engineering perspective. Led transition to an open source software model in 2021, increasing adoption across commercial and government clients and reducing implementation friction. Fulfilled numerous federal government research and development contracts in the education and simulation space as Principal Investigator (PI).

Serve on related IEEE committees and working groups, shaping global learning data standards (LTSC Leadership, P2881, P2997, 9274) and acting as liaison between the IEEE Learning Technology Standards Committee (LTSC) and Simulation Interoperability Standards Organization (SISO).

Primary technologies include Java, Clojure, Javascript/Typescript, React, Python, PHP, PostgreSQL, GraphQL, Datomic, and xAPI. Deploy systems on AWS, Azure and custom / government deployment environments. Heavy emphasis on Infrastructure as Code, particularly Cloudformation. Maintain secured hardened DoD versions of software in high security environments. Deployed custom ML algorithms for event analysis and training content recommendations.

Notable clients/projects have included the US DoD Advanced Distributed Learning Initiative, US Navy NETC, US Air Force, US Space Force, US Army STTC, Deloitte, Boeing, IBM, HTX, Relias, Moodle, NYU, Verizon, American Red Cross, the New Zealand Defence Force, the Netherlands Armed Forces.

Chief Technology Officer

SafetyPIN Technologies Inc, Baltimore MD | Jan 2018-Sep 2019

Architected and built the core systems for a B2C startup in the background check and ID verification space. Responsibilities included development and maintenance of the core product, technical platform and vendor selection and integration for marketing and peripheral technologies, hiring and managing development and QA resources, contractor and vendor management, enterprise partner integration, code reviews, security analysis, and directing and advising the company from a technical perspective.

Core technologies included Java, SQL, Javascript/AngularJS, and a heavy reliance on AWS IaaS services such as EC2, RDB, ElastiCache, SQS and SES. Designed and developed a machine learning classification model which automated parts of background check application processing, improving processing speed and consistency. Developed mobile companion applications for Android and iOS.

Principal Engineer

South Ferry IT Consulting LLC, Baltimore MD | Nov 2012– Present

Created a consulting company in 2010 to execute projects for various clients and media agencies in roles such as Lead Developer, Technical Director, Advisor, and Interim CTO. Built applications for client needs across many platforms and industries. Clients have included Norfolk Southern, Washington Theological Consortium, Colorado Donors Alliance, Staffing Advisors, Adboku, Interfaith Conference of Washington, IRN, Catholic Volunteers Network, Arbit Inc, StudioRHM, Chinese Language Institute, Perisphere Media, and the University of California Riverside.

Chief Solutions Architect

Publishing Services, Agora Inc, Baltimore MD | Dec 2013- Dec 2017

Served as the senior technical authority across the organization's enterprise technology portfolio, reviewing technology choices, driving modernization of legacy systems, and designing and developing new systems. Architected solutions for complex technical problems spanning multiple products and led integration efforts across the enterprise stack.

The organization built and maintained ecommerce and marketing systems for 25+ multinational publishing companies through five primary software projects, processing transactions for tens of millions of customers and hundreds of millions of dollars in revenue annually. When complex problems arose, worked directly with a small team of senior engineers to diagnose, architect, and implement solutions, including system scaling, integration, and modernization.

Lead Engineer (E-Commerce)

Publishing Services, Agora Inc, Baltimore MD | Jan 2012– Dec 2013

Led a team of Java developers to build and expand a high-volume multinational e-commerce payments platform, and provide direction on many projects for a publishing consortium. Transitioned the entire engineering team from consultants to in-house developers through focused hiring and training. The role included system architecture, senior software

engineering responsibilities, and the hiring, training and guidance of many junior and mid-level developers. The role also consisted of code reviews, coordinating sensitive production deployments and migrations, and communicating effectively with product owners, project managers, QA, support and clients.

Lead Application Developer

Government Services IPT, Largo MD | Jan 2010–Jan 2012

Worked as a software engineer for a government contractor building applications for Federal / DoD clients across a wide range of requirements and platforms. Built Project Management / ERP solutions, EVM tools and FCA (Facility Condition Assessment) software. Core technologies included Java, PHP, Javascript, MySQL, PostgreSQL. Worked as part of a small SCRUM team and integrated agile development methodologies in day-to-day work. Notable clients included NASA and the US Army Corps of Engineers (USACE).

Developer / Wireless Systems Planner

NCS, McGill University, Montreal QC, Canada | May 2008–July 2009

Started as a deployment and planning specialist designing an updated wireless service infrastructure across numerous campuses. Transitioned into applications development and worked in an MRTG / SNMP / PHP / Perl / Bash environment writing applications to provide network data for real-time monitoring of tens of thousands of devices across campus networks in the Montreal area. Job was performed while in pursuit of a B. Engineering.

Application Developer

Antharia LLC, Seabrook MD | June 2001–Dec 2007

Software developer for a web media studio, contributing to a diverse line of browser-based software products using PHP, MySQL and Js. Products included a comprehensive custom CMS, CRM, and payment and mailing tools and the clients consisted primarily of NPOs in the DC area. The position fluctuated between part-time and full-time based on education commitments. Some notable clients included the ACLU and Discovery Channel.

EDUCATION

Bachelor of Engineering (Computer)

McGill University, Montreal, QC | 2009

(Note: U.S. Citizen)

PUBLICATIONS

*Morse, K., Honold, E., Casey, C., Riley, B. (2026). **Integrating High Level Architecture (HLA) Federation Object Model (FOM) and Experience API (xAPI)**. Proceedings of the 2026 Simulation Interoperability Standards Organization Simulation Innovation Workshop. Orlando, FL.*

*Robson, R., Goldberg, B., Blake-Plock, S., Casey, C., Hoyt, W., Hernandez, M., Ray, F. (2022). **Mining Artificially Generated Data to Estimate Competency**. Proceedings of the 15th International Conference on Educational Data Mining. Durham University. Durham, UK.*

*Owens, K., Goldberg, B., Robson, R., Hoffman, M., Ray, F., Colburn, A., Hernandez, M., Blake-Plock, S., Casey, C. (2022). **Synthetic Training Environment Experiential Learning for Readiness (STEEL-R)**. Proceedings of the Tenth Annual GIFT Users Symposium. US Army Combat Capabilities Development Command - Soldier Center. Orlando, FL.*

Blake-Plock, S., Hoyt, W., & Casey, C. (2021). **Instrumenting GIFT with xAPI: a use case for IEEE P9274.3.x standards activity and implications for the broader field of ITS and AIS**. *Proceedings of the Ninth Annual GIFT Users Symposium*. US Army Combat Capabilities Development Command - Soldier Center. Orlando, FL.

Blake-Plock, S., Hoyt, W., Casey, C., & Zapata-Rivera, D. (2020). **Data Analytics and Visualization for xAPI Learning Data: Considerations for a GIFT Strategy**. In Sinatra, A.M., Graesser, A.C., Hu, X., Goldberg, B., and Hampton, A.J. (Eds.). *Design Recommendations for Intelligent Tutoring Systems: Volume 8 - Data Visualization*. US Army Combat Capabilities Development Command - Soldier Center. Orlando, FL.

Blake-Plock, S. and Casey, C. (2020). **DATASIM and xAPI Profile Validation**. *iFEST 2020*. National Training and Simulation Association. Alexandria, VA.

OPEN SOURCE CONTRIBUTIONS

[SQL LRS](#) - Clojure

Apache 2.0 Open Source Learning Record Store (LRS) with support for SQLite, PostgreSQL and MySQL/MariaDB. Created this product in 2020 to pivot Yet Analytics to an Open Source software model, led Yet engineering team in developing it, and it became one of the most widely used LRS implementations in the world.

[Federate xAPI](#) - Java

Apache 2.0 Open Source High Level Architecture (HLA) Federate for transforming simulation data into xAPI. Developed the idea for this product and the initial prototype while serving as the liaison between the IEEE LTSC and SISO. Led the team to bring the project to maturity in 2026.

[Enterprise Learner Record Repository \(ELRR\)](#) - Java/Spring

Apache 2.0 Open Source IEEE P2997 Learner Record Repository. Developed in collaboration with US DoD and Deloitte.

[Java xAPI Tools](#) - Java

Small Apache 2.0 OSS library for handling xAPI data in Java, complete with conformant client. Developed client to fill a need in a project and continue to maintain and publish it on Maven Central.

[DATASIM](#) - Clojure

Apache 2.0 Open Source Synthetic xAPI Data Generator. Developed in collaboration with US DoD.

[LaneGraph](#) - C++ (Unreal)

System for generating virtual n-lane traffic corridors from splines which heavily influences pathfinding algorithms without restricting movement. In practice this makes simulated traffic seem more organic and able to deal with emergent obstacles. Includes custom A* nav implementation integrated with traffic corridors. Created out of necessity for a hobby project.

[SplintFormation](#) - C++ (Unreal)

Toolkit primarily for extracting and manipulating information from Unreal Landscape Splines which are locked down and absent a robust API in Unreal. Was used for procedurally generated buildings and other content along a landscape-created spline paths/roads and includes some helpers for algorithmically packing splines with stacked multi-length objects. Created out of necessity for a hobby project.

[Moodle xAPI Logstore](#) - PHP

Contributed to a 2024 overhaul of a widely used xAPI implementation for one of the most popular Learning Management Systems, enabling comprehensive xAPI instrumentation.